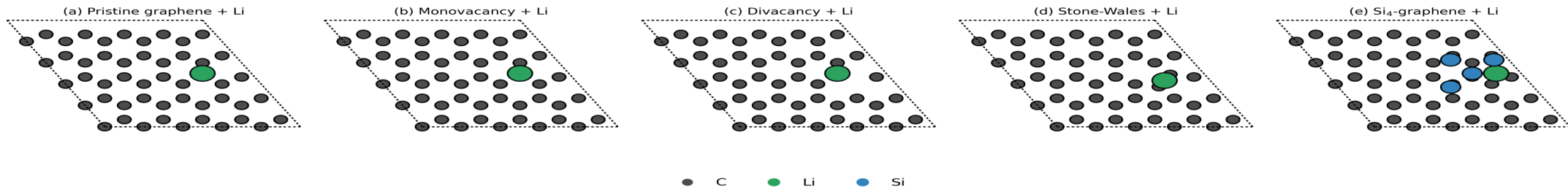


Validation-first AI for local lithium energetics

DFT anchors | leakage-aware MACE evaluation | out-of-domain stress tests



1 First-principles anchors

273

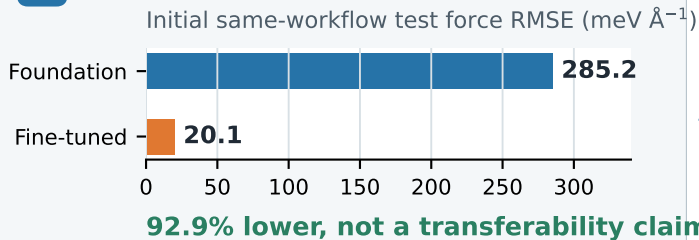
DFT-labeled MLIP frames

PBE-D3(BJ)/dipole adsorption

-3.349 to 0.009 eV/Li

194 relaxation + 79 site/path frames

2 Audit before scale-up



Group-aware split | E_0 calibration | 3 model seeds

3 Gate physical claims

SUPPORTED

Local adsorption screening

Model-domain diagnosis

WITHHELD

Migration barriers

Diffusion coefficients

16 converged DFT snapshots; 9 MACE-vs-DFT force tests